William Pang

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Education

University of Waterloo September 2025 – May 2030
Bachelor of Applied Science in Computer Engineering *Waterloo, ON*

Technical Skills

Languages: TypeScript/JavaScript, Python, C#, C/C++, Java, Go, HTML/CSS
Frameworks: Angular, .NET, FastAPI, React, Next.js, Node.js, Gin
Cloud & Database: AWS, PostgreSQL, Redis, SQLite
Testing & DevOps: Git, Jest, Playwright, Docker, CI/CD

Work Experience

Software Developer Intern September 2026 – December 2026
Tangam Systems | TypeScript, Angular, C#, .NET *Waterloo, ON*

- Developed a table spread planning feature end to end, including **C# APIs** that pull historical table occupancy and win data and an **Angular** for comparing spread scenarios for **300+** properties
- Migrated TYM's table allocation page from ASPX to modern **.NET**, improving page load time by **30%**
- Rearchitched the deprecated Karma test runner with Vitest across a 1,800-test **Angular** codebase, reducing CI pipeline and developer test time by **40%**

Software Developer Intern January 2026 – April 2026
Port 443 Inc. | TypeScript, Angular, C#, .NET, Playwright *Burlington, ON*

- Shipped a mileage-tracking feature in TypeScript for a platform serving **30,000 users**, iterating post-launch through customer feedback and monitoring
- Designed a robust error-handling system in **C#** and **.NET** framework, reducing unhandled exceptions and improving application stability in production
- Identified testing gaps and designed **200+** rigorous automated tests with **Jest** and **Playwright** to validate business logic and UI behavior, reducing regression bugs by **28%** over three release cycles

Ground Station Developer March 2026 – August 2026
UW Orbital | Python, FastAPI, Pydantic, python-jose *Waterloo, ON*

- Redesigned the authentication token system in **Python** by implementing sliding-session expiration logic, improving token persistence and automated session renewal for authenticated users
- Implemented **Pydantic** validation and filtering logic to parse incoming CubeSat sensor readings

Projects

HomeMatch | *Python, PyTorch, FastAPI, PostgreSQL, React* July 2026

- Built an ML-powered sublet matching platform for University of Waterloo students with a **Python** backend, scraping and aggregating **300+** listings from 5 sources into a single searchable feed
- Developed a preference-based ranking model using cosine similarity over 12 features (price, distance to campus, term dates, roommates), reaching **87%** top-5 match relevance

Auction Bid Processing System | *Go, Gin, Redis, PostgreSQL, Docker* September 2026

- Designed and exposed auction creation and bid submission endpoints with **Go** and **Gin**
- Used goroutines and **Redis** queues to implement concurrent bid processing that handles **75+ bids** per second while ensuring the highest bidder wins
- Designed a PostgreSQL database schema with locking and validation logic to ensure invalid bids are excluded

Texas Hold'em Game Engine | *C++* May 2026

- Architected the engine using **OOP** principles with separate Deck, Hand, Player, Table, and GameManager classes, enabling modular testing and extensibility
- Designed a state machine architecture to manage game phases with clean transitions and robust edge-case handling
- Optimized hand evaluation using precomputed rank lookup tables, reducing per-hand comparison latency by **95%**